

B Correspondence to the NGCC Security Strength Levels

NGCC requires submitted algorithms to support classical security strength levels of 128, 256, and 512 bits and states that “The corresponding quantum-resistant security strength shall be no less than 80, 128, and 256 bits.”¹ Accordingly, the SYDO-160, SYDO-256, and SYDO-512 parameter sets correspond, respectively, to the three required NGCC levels. The SYDO-160 parameter set targets 160-bit classical security and 80-bit quantum security, and therefore exceeds the 128-bit classical requirement while meeting the 80-bit quantum-resistant requirement of the first level; SYDO-256 targets 256-bit classical security and 128-bit quantum security, matching the second level; and SYDO-512 targets 512-bit classical security and 256-bit quantum security, matching the third level.

¹ Institute of Commercial Cryptography Standards, *Submission Requirements for Public-Key Cryptographic Algorithms*, Section 2(1), October 2025. Available at: <https://www.niccs.org.cn/niccs/Notice/1975896137741635584/IDop1mav.pdf>.